The RHF and CASSCF analytic nuclear Hessians

design notes: derivation, coupled-perturbed equations, density fitting and the state average

free-breakfast design notes

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Abstract

Part I is the closed-shell RHF Hessian: the derivation in the vocabulary of the tree's RHF gradient, two equivalent forms of the result (one manifestly symmetric, one made only of AO matrices and Fock builds), the CPHF equations, the skeleton Fock derivatives every right-hand side needs, the density-fitted variant, the algorithm, its checks and a build order. By Wigner's $2n + 1$ rule no second-order response is needed. Part II is the CASSCF Hessian, state-specific and state-averaged: the second-order connection, the CP-CASSCF right-hand side and its $3N$ solves with the Z-vector's operator, the state average's Lagrangian, whose second derivative needs the gradient's own Z and a third derivative of the averaged energy contracted once with it, the rotations among the averaged roots, the explicit part over density fitting, the algorithm, the conditions it needs, its checks and a build order.

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Part I
RHF

1 Notation and the energy

Orbitals are real and the reference is closed-shell RHF with n_o doubly occupied orbitals. AO indices are $\mu\nu\lambda\sigma$, occupied MOs $ijkl$, virtual MOs $abcd$, and general MOs $pqrs$. C is the MO coefficient matrix, C_o and C_v its occupied and virtual columns, and ε_p the canonical orbital energies. Nuclear coordinates are x and y; each is one Cartesian component of one nucleus, $x = (A, k)$, and there are $3N$ of them for N atoms.

The density and energy-weighted density are

$$D = 2\,C_o C_o^{\mathsf{T}}, \qquad W = 2\,C_o\,\varepsilon_o\,C_o^{\mathsf{T}} = \tfrac{1}{2}\,DFD, \tag{1}$$

the second form of W holding for any S-orthonormal C that diagonalizes the occupied block of F (since $C_o^{\mathsf{T}} F C_o = \varepsilon_o$). It is the form that differentiates cleanly.

With the antisymmetrized integral

$$(\mu\nu\|\lambda\sigma) = (\mu\nu|\lambda\sigma) - \tfrac{1}{2}(\mu\lambda|\nu\sigma), \tag{2}$$

which is symmetric under $(\mu\nu) \leftrightarrow (\lambda\sigma)$ for real functions, the two-electron operator on a symmetric density P is

$$G(P)_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma}(\mu\nu\|\lambda\sigma) = J(P)_{\mu\nu} - \tfrac{1}{2}K(P)_{\mu\nu}, \tag{3}$$

and G is a symmetric bilinear form: $\mathrm{tr}[P\,G(Q)] = \mathrm{tr}[Q\,G(P)]$. The Fock matrix is $F = h + G(D)$ and the energy is

$$E = \sum_{\mu\nu} D_{\mu\nu}h_{\mu\nu} + \tfrac{1}{2}\sum_{\mu\nu\lambda\sigma} D_{\mu\nu}D_{\lambda\sigma}(\mu\nu\|\lambda\sigma) + V_{nn}. \tag{4}$$

The SCF conditions are $F_{ai} = 0$ and $C^{\mathsf{T}}SC = 1$, at every geometry.

Derivative integrals. A superscript on an integral means its derivative at fixed AO coefficients, the basis functions moving with their nuclei: $h^x_{\mu\nu} = \partial h_{\mu\nu}/\partial x$, S^x, and $(\mu\nu|\lambda\sigma)^x$; likewise h^{xy}, S^{xy} and $(\mu\nu|\lambda\sigma)^{xy}$. h^x includes the derivative of the nuclear-attraction *operator* as well as of the basis functions. MO transforms with the unperturbed coefficients are written $S^x_{pq} = C_p^{\mathsf{T}}S^xC_q$, and similarly for any AO matrix.

Skeleton Fock derivative. The derivative of F at fixed D,

$$F^{(x)}_{\mu\nu} = h^x_{\mu\nu} + \sum_{\lambda\sigma} D_{\lambda\sigma}(\mu\nu\|\lambda\sigma)^x, \tag{5}$$

is the central intermediate. There are $3N$ of them, each an AO matrix. The Hessian is an explicit part, the second-derivative integrals contracted with the converged density, plus a response part that needs the first-order orbital response, one coupled-perturbed Hartree–Fock (CPHF) solve per nuclear coordinate; $F^{(x)}$ enters both, and every CPHF right-hand side.

2 The gradient, as the tree computes it

Differentiating (4) and using the SCF conditions gives the familiar

$$E^x = \sum D_{\mu\nu}h^x_{\mu\nu} + \tfrac{1}{2}\sum D_{\mu\nu}D_{\lambda\sigma}(\mu\nu\|\lambda\sigma)^x - \sum W_{\mu\nu}S^x_{\mu\nu} + V^x_{nn}, \tag{6}$$

which is exactly `gradient::rhf`'s term list (`src/gradient/README.md`): `kinetic_gradient` and `potential_gradient` at D, `overlap_gradient` at $-W$, `coulomb_gradient` on D and $-\frac{1}{2}$ `exchange_gradient` on D, and the nuclear repulsion. The orbital response is absent because E is stationary in the occupied–virtual rotations, and the normalization response is carried by $-WS^x$.

The Hessian is the derivative of (6). Equation (6) holds at *every* geometry provided D and W are the converged ones there, so differentiating it is legitimate, and two kinds of term result: those where y hits an integral (the *explicit* part) and those where y hits D or W (the *response* part):

$$E^{xy} = E^{xy}_{\mathrm{expl}} + \sum D^y_{\mu\nu}\,F^{(x)}_{\mu\nu} - \sum W^y_{\mu\nu}\,S^x_{\mu\nu}, \tag{7}$$

$$E^{xy}_{\mathrm{expl}} = \sum D_{\mu\nu}h^{xy}_{\mu\nu} + \tfrac{1}{2}\sum D_{\mu\nu}D_{\lambda\sigma}(\mu\nu\|\lambda\sigma)^{xy} - \sum W_{\mu\nu}S^{xy}_{\mu\nu} + V^{xy}_{nn}. \tag{8}$$

The middle term of (7) collects D^yh^x and the two halves of $\frac{1}{2}(DD)^y(\|)^x$, which are equal by the pair symmetry of $(\|)$. Here D^y and W^y are *total* derivatives: they need the orbitals' response.

3 The orbital response

3.1 Parametrization and the orthonormality constraint

Expand the perturbed coefficients in the unperturbed ones,

$$\frac{\partial C_{\mu p}}{\partial y} = \sum_q C_{\mu q}\, U^y_{qp}. \tag{9}$$

Differentiating $C^\mathsf{T} S C = 1$ gives

$$U^y_{pq} + U^y_{qp} + S^y_{pq} = 0. \tag{10}$$

The occupied–occupied block of U^y is a rotation among occupied orbitals plus a symmetric part fixed by (10). D and E are invariant to the rotation, so we choose

$$U^y_{ij} = -\tfrac{1}{2} S^y_{ij} \tag{11}$$

(the orbitals do *not* stay canonical along y; this is why W is better differentiated as $\frac{1}{2}DFD$ than as $2C_o\varepsilon_o C_o^\mathsf{T}$). The virtual–occupied block U^y_{ai} is the unknown, and the occupied–virtual block follows from (10): $U^y_{ia} = -U^y_{ai} - S^y_{ai}$. The virtual–virtual block never enters a closed-shell quantity.

3.2 The density derivative

With (11),

$$D^y = D^y_U - D^y_S, \qquad D^y_U = 2\sum_{ai} U^y_{ai}\,(C_a C_i^\mathsf{T} + C_i C_a^\mathsf{T}), \qquad D^y_S = 2\, C_o\, S^y_{oo}\, C_o^\mathsf{T}, \tag{12}$$

where S^y_{oo} is the $n_o \times n_o$ block of S^y_{pq}. D^y_S is known from the overlap derivative alone; D^y_U needs the CPHF solution. In code $D^y_U = 2(X + X^\mathsf{T})$ with $X = C_v U^y C_o^\mathsf{T}$: two GEMMs.

3.3 The coupled-perturbed Hartree–Fock equations

Differentiate the condition $F_{ai} = C_a^\mathsf{T} F C_i = 0$. Three pieces appear: the coefficient derivatives on either side, the skeleton derivative (5), and the Fock change from D^y:

$$0 = \frac{dF_{ai}}{dy} = U^y_{ia}\,\varepsilon_i + U^y_{ai}\,\varepsilon_a + F^{(y)}_{ai} + G(D^y)_{ai}. \tag{13}$$

Substituting $U^y_{ia} = -U^y_{ai} - S^y_{ai}$ and (12):

$$\boxed{\sum_{bk} M_{ai,bk}\, U^y_{bk} = B^y_{ai}, \qquad B^y_{ai} = -F^{(y)}_{ai} + \varepsilon_i\, S^y_{ai} + G(D^y_S)_{ai}} \tag{14}$$

with the electronic Hessian

$$M_{ai,bk} = (\varepsilon_a - \varepsilon_i)\,\delta_{ab}\delta_{ik} + 4(ai|bk) - (ab|ik) - (ak|ib). \tag{15}$$

M is symmetric, and positive definite exactly when the RHF solution is a minimum under real singlet rotations (it is the real RHF→RHF stability matrix up to a factor). A CPHF solve that does not converge is therefore a diagnostic of an unstable reference, not of the solver.

The product never forms M. For a trial U,

$$(MU)_{ai} = (\varepsilon_a - \varepsilon_i)\, U_{ai} + C_a^{\mathsf{T}}\, G(D_U)\, C_i, \tag{16}$$

with D_U built from U as in (12). So one product is two GEMMs to D_U, one J/K build on a symmetric density, and two GEMMs back to the ai block: the same work as one SCF iteration. The $3N$ right-hand sides are independent, so the J/K builds batch across them, and the diagonal $(\varepsilon_a - \varepsilon_i)$ is the obvious preconditioner.

How many solves. One per coordinate, $3N$, or by translational invariance $3(N-1)$: the response to moving every nucleus together is a pure translation of the orbitals, so one atom's three responses are minus the sum of the others'. Rotational invariance gives three more only at a stationary point and only with care; it is not worth it. The $2n+1$ rule says first-order response suffices for a second derivative: no U^{xy} is ever needed. The Z-vector trick does not help here: it removes response solves when there is one functional to differentiate, and a Hessian has $3N$ gradient components.

Since written. The CPHF is `response::cphf` (`src/response/cphf`), every right-hand side in one block.

4 The Hessian

4.1 The AO form

Equation (7) with D^y from (12) and W^y from (1),

$$W^y = \tfrac{1}{2}(D^y F D + D F^y D + D F D^y), \qquad F^y = F^{(y)} + G(D^y), \tag{17}$$

is already a complete algorithm: per coordinate y, form D^y, one J/K build for $G(D^y)$, a handful of GEMMs for W^y, then $3N$ dot products $\mathrm{tr}(D^y F^{(x)}) - \mathrm{tr}(W^y S^x)$ for every x. It is not manifestly symmetric in x and y, but it is the exact derivative of an exact gradient, so its asymmetry measures the CPHF residual. Symmetrizing $\tfrac{1}{2}(E^{xy} + E^{yx})$ at the end is standard, and the discarded antisymmetric part is a free convergence diagnostic.

4.2 The symmetric form

Expanding (7) in MOs, using (14) to eliminate the $F_{ai}^{(x)}$ and the bilinear symmetry of G (two terms $\pm 4\sum_{ai} U_{ai}^y\, G(D_S^x)_{ai}$ cancel, one from each of D^y and W^y), gives

$$\boxed{\begin{aligned} E^{xy} = E^{xy}_{\mathrm{expl}} &- 4\sum_{ai} B_{ai}^x\, U_{ai}^y - 2\sum_{ij} (S_{ij}^x\, F_{ij}^{(y)} + S_{ij}^y\, F_{ij}^{(x)}) \\ &+ 2\sum_{ij} S_{ij}^x\, S_{ij}^y\, (\varepsilon_i + \varepsilon_j) + \mathrm{tr}\big[D_S^x\, G(D_S^y)\big]. \end{aligned}} \tag{18}$$

Every term is symmetric in $x \leftrightarrow y$; the first because $\sum_{ai} B_{ai}^x U_{ai}^y = B^{x\mathsf{T}} M^{-1} B^y$. That term is the orbital relaxation, and it is negative semi-definite since M is positive definite: letting the orbitals relax can only lower the curvature. It is the term that a frozen-orbital Hessian omits and that dominates the error of one.

The two forms agree to the CPHF tolerance (section 9). (18) costs fewer J/K builds (the $G(D_S^y)$ are needed for B^y anyway) and needs no W^y; (7) is shorter to write over AO matrices. The recommendation is (18), with (7) kept as a test.

4.3 Nuclear repulsion

With $\mathbf{R} = \mathbf{R}_A - \mathbf{R}_B$ and $R = |\mathbf{R}|$, the off-diagonal blocks are

$$\frac{\partial^2 V_{nn}}{\partial A_k \, \partial B_l} = -Z_A Z_B \, \frac{3R_k R_l - R^2 \delta_{kl}}{R^5} \qquad (A \neq B), \tag{19}$$

and each diagonal block is minus the sum of its row's off-diagonal blocks.

Since written. The response part, in the symmetric form (18), is `hessian::df_rhf_response` (`src/hessian`), and its tests build the AO form (7) from the same U^x as the check this section recommends.

5 The explicit part and its integrals

E^{xy}_{expl} (8) needs four second-derivative integral classes, each only ever contracted, never stored:

class	contracted with	centres differentiated
S^{xy}	$-W$	2 (bra, ket)
T^{xy}	D	2
V^{xy}	D	2, plus every nucleus C (operator)
$(\mu\nu\|\lambda\sigma)^{xy}$	$\frac{1}{2} D \otimes D$ (antisymmetrized)	4

Shape of the output. A shell pair touches two atoms, so an overlap or kinetic second derivative fills at most a 6×6 block of the $3N \times 3N$ Hessian, and a nuclear-attraction one, whose operator nucleus C is a third atom, a 9×9 block; a quartet touches four, a 12×12 block with 78 unique entries. Translational invariance, $\sum_A \partial/\partial A_k = 0$ applied to each integral, removes one centre: a quartet's derivatives need only three centres' worth of distinct integrals (45 unique second-derivative entries), and an overlap or kinetic pair's only one. The tree's first-derivative kernels do not use translational invariance (`src/integrals/one_electron/gradient/README.md`); for second derivatives the saving is large enough that they should.

The nuclear-attraction operator. V^{xy} has three kinds of term: both derivatives on basis centres, one on a basis centre and one on the operator's nucleus, and both on the operator's nucleus. The last is the second derivative of the electrostatic potential at nucleus C, the same integral family as electric-field-gradient integrals. Translational invariance turns the operator derivatives into basis-centre ones, $\partial_C = -(\partial_A + \partial_B)$ for the C-term of a pair on A, B, which is how most codes avoid a separate kernel.

Angular momentum. In McMurchie–Davidson form a derivative on a Gaussian is a sum of functions of $l \pm 1$, so a second derivative needs Hermite expansions to $l + 2$ on that centre, and a quartet's to $L + 2$ in total over the two centres differentiated. At f this is the same register and stack pressure that bounded the f-shell first-derivative and exchange kernels; the generated kernels' compile time and spill behaviour should be measured with `devtools/kernel-compile-watchdog.sh` before anything is assumed.

Since written. The one-electron contractions are `overlap_hessian`, `kinetic_hessian` and `potential_hessian` (`src/integrals/one_electron/hessian`), their diagonal blocks from translational invariance; the fitted two-electron ones are `metric_hessian` and `three_center_hessian` (`src/integrals/eri/density_fitting/hessian`).

6 The skeleton Fock derivatives

Both forms of the Hessian, and every CPHF right-hand side, need the $3N$ AO matrices $F^{(x)}$ (5). The two-electron part is a *derivative J/K build*: $J^{(x)}(D)_{\mu\nu} = \sum D_{\lambda\sigma}(\mu\nu|\lambda\sigma)^x$ and the matching $K^{(x)}$, for every x at once. This is the kernel the tree most clearly lacks. Its first-derivative integrals exist (`coulomb_gradient` and `exchange_gradient` generate $(\mu\nu|\lambda\sigma)^x$), but they contract both index pairs with D and return $3N$ scalars. A Hessian needs the same integrals contracted with one D, returning $3N$ matrices: $3N\,n_{\mathrm{AO}}^2$ doubles of output, and scatter writes into per-atom slabs.

The one-electron part. h^x as matrices rather than scalars is the same change to `kinetic_gradient` and `potential_gradient`, and S^x as matrices to `overlap_gradient`; all three are cheap.

A shell quartet differentiated on the centre of shell μ contributes to $J^{(x)}$ through every $(\mu\nu)$ and $(\lambda\sigma)$ element it touches, so the natural layout is one n_{AO}^2 slab per coordinate, written atomically or reduced per block. With translational invariance, three of four centres suffice and the fourth's contribution is minus the sum. Memory: at 500 AOs and 30 atoms the $F^{(x)}$ alone are 180 MB, and the CPHF's responses and right-hand sides are small beside them.

Since written. The one-electron part is `overlap_derivative`, `kinetic_derivative` and `potential_derivative`, the $[n_{\mathrm{atom}}, 3, n_{\mathrm{AO}}, n_{\mathrm{AO}}]$ tensors of `src/integrals/one_electron/gradient`; the two-electron part, over density fitting, is `df_jk_derivative` (`src/integrals/eri/density_fitting/gradient`). `src/hessian` assembles $F^{(x)}$ from them.

7 Density fitting

The tree's DF-RHF gradient fits J and K with a Coulomb-metric auxiliary basis, $M_{PQ} = (P|Q) = (LL^{\mathsf{T}})_{PQ}$ and $T_{\mu\nu,P} = (\mu\nu|P)$. The DF Hessian has the same structure as sections 3 and 4, with G replaced by the DF G everywhere (in F, in the CPHF product (16), and in $G(D_S^y)$), and the explicit and skeleton parts differentiated through the fit. The Coulomb-metric fit is variational in the fit coefficients, so no fit response appears; only M^{-1}'s derivative does.

7.1 Coulomb

$E_J = \frac{1}{2}\,u^{\mathsf{T}}M^{-1}u$ with $u = T^{\mathsf{T}}\mathrm{vec}\,D$ and $d = M^{-1}u$, the vector `gradient::df_rhf` already builds. At fixed D,

$$E_J^x = u^{x\mathsf{T}}d - \tfrac{1}{2}\,d^{\mathsf{T}}M^x d, \tag{20}$$

$$E_J^{xy} = u^{xy\mathsf{T}}d - \tfrac{1}{2}\,d^{\mathsf{T}}M^{xy}d + \left(u^x - M^x d\right)^{\mathsf{T}}M^{-1}\left(u^y - M^y d\right), \tag{21}$$

with $u^x = T^{x\mathsf{T}}\mathrm{vec}\,D$. The last term is new relative to the gradient: a $3N \times 3N$ Gram matrix of the vectors $L^{-1}(u^x - M^x d)$, each of length n_{aux}: one triangular solve per coordinate and one GEMM.

7.2 Exchange

$E_x = -\sum_{ij} T_{ij}^{\mathsf{T}} M^{-1} T_{ij}$ with $T_{ij,P} = (ij|P)$ at fixed MO coefficients and $c_{ij} = M^{-1}T_{ij}$, the fitted occupied pairs of the gradient. At fixed C,

$$E_x^x = -\sum_{ij} (2\, T_{ij}^{x\mathsf{T}} c_{ij} - c_{ij}^{\mathsf{T}} M^x c_{ij}), \tag{22}$$

$$E_x^{xy} = -\sum_{ij} \Big(2\, T_{ij}^{xy\mathsf{T}} c_{ij} - c_{ij}^{\mathsf{T}} M^{xy} c_{ij} + 2\,(T_{ij}^x - M^x c_{ij})^{\mathsf{T}} M^{-1} (T_{ij}^y - M^y c_{ij})\Big). \tag{23}$$

The cross term is the DF Hessian's bottleneck: it needs the half-transformed derivative three-centre integrals $T_{ij,P}^x$ for every coordinate, $3N\, n_o^2\, n_{\text{aux}}$ doubles in all. It should be built per atom (only that atom's AOs and aux functions contribute to T^x) and contracted into the Hessian one pair of atom blocks at a time, never held whole.

These are the fixed-orbital (explicit) parts. Their skeleton analogues, the derivative of the DF Fock matrix at fixed D, follow the same way: $J^{(x)} = T^x d + T\, M^{-1}(u^x - M^x d)$, and the exchange analogue with c_{ij}.

7.3 What the fit adds to the kernel list

$(\mu\nu|P)^{xy}$ and $(P|Q)^{xy}$, the second-derivative three- and two-centre integrals, contracted with $D \otimes d$, $d \otimes d$ and the exchange analogues of `gradient::df_rhf`'s Γ^K and W^K; and $(\mu\nu|P)^x$ and $(P|Q)^x$ as *uncontracted* per-coordinate slabs (or contracted over one index only) for (21), (23) and the skeleton Fock. The existing `three_center_gradient` and `metric_gradient` compute the right integrals and contract them to scalars, like their direct counterparts. The aux side carries shells two l above the orbital side (to i), so second derivatives push the aux Hermite expansion to $l = 8$; that is a strong argument for using translational invariance to move the derivatives off the aux centre wherever possible.

Since written. The two-electron explicit part, (21) and (23) with their cross terms built one atom block at a time, is `hessian::df_rhf_two_electron` (`src/hessian`).

8 The algorithm

Cost. The J/K builds dominate: $3N$ for the $G(D_S^x)$, and $3N$ per CG iteration, typically 10–20 iterations to 10^{-8} in the residual. That is $3N$ times a short SCF. The derivative J/K build for the $F^{(x)}$ costs a small multiple of one gradient. The second-derivative ERI contraction is one pass over the quartets, more expensive per quartet than a gradient but done once. For a molecule of 30 atoms the CPHF is about 90 batched solves, and it is the term to make fast. (These are expectations from the operation counts, not measurements.)

The CPHF solver. All $3N$ systems share M, so a block CG (or plain CG run in lockstep, with per-column convergence masks) turns the $3N$ J/K builds per iteration into one batched build over $3N$ densities. The integral-direct J/K and the DF J/K both take one density per call today; a multi-density build is the natural extension, and it serves TDHF and polarizabilities too. Memory per column is $n_v n_o$ for each of U, B, the residual and the search direction: small. `response::zvector` is the tree's closest existing linear solver over a response product and is the model for its interface and stopping rules.

Algorithm 1 RHF analytic Hessian (symmetric form (18))

Input: converged C, ε, n_o; D, W; the basis (and aux basis)
Output: E^{xy}, $3N \times 3N$

$H \leftarrow E^{xy}_{\text{expl}}$: second-derivative integrals contracted with D, $-W$, $\frac{1}{2} D \otimes D$ (8); add V^{xy}_{nn}
for each coordinate x (batched) **do**
    S^x, $F^{(x)}$ as AO matrices ▷ derivative J/K build
    $S^x_{pq} \leftarrow C^\mathsf{T} S^x C$, $F^{(x)}_{pq} \leftarrow C^\mathsf{T} F^{(x)} C$ (occupied columns only)
    $D^x_S \leftarrow 2 C_o S^x_{oo} C_o^\mathsf{T}$, then $G(D^x_S)$ ▷ one J/K build
    $B^x_{ai} \leftarrow -F^{(x)}_{ai} + \varepsilon_i S^x_{ai} + G(D^x_S)_{ai}$
solve $MU^x = B^x$ for all x together by preconditioned CG (16), preconditioner $(\varepsilon_a - \varepsilon_i)^{-1}$
$H \mathrel{+}= -4\, B^\mathsf{T} U - 2(\Sigma + \Sigma^\mathsf{T}) + 2\,\Omega + \Theta$, where $\Sigma_{xy} = \sum_{ij} S^x_{ij} F^{(y)}_{ij}$, $\Omega_{xy} = \sum_{ij} S^x_{ij} S^y_{ij} (\varepsilon_i + \varepsilon_j)$, $\Theta_{xy} = \mathrm{tr}[D^x_S G(D^y_S)]$
check $\|H - H^\mathsf{T}\|$ and the translational sum rule; symmetrize

Since written. `hessian::df_rhf` is Algorithm 1 end to end over density fitting (`src/hessian/README.md`, "The whole Hessian").

9 Checks

Formulas. The design implemented (14), (7) and (18) with PySCF supplying the integrals (E^{xy}_{expl} from `partial_hess_elec` plus `hess_nuc`, $F^{(x)}$ from `make_h1`, S^x from `int1e_ipovlp`) and solved the CPHF directly with a dense M. For water in 6-31G at a distorted C_1 geometry:

comparison	max. abs. difference (E_h/a_0^2)
symmetric form (18) vs. `pyscf.hessian.rhf`	1.9×10^{-8}
symmetric form vs. central difference of analytic gradients	7.0×10^{-8}
AO form (7) vs. `pyscf.hessian.rhf`	1.9×10^{-8}
asymmetry of the AO form	3.9×10^{-10}
asymmetry of the symmetric form	1.5×10^{-15}

The largest Hessian element is 0.71, so the agreement is at PySCF's own CPHF tolerance and the finite difference's step error.

For the GPU implementation. In order of how cheaply they catch a bug:

1. $H = H^\mathsf{T}$ in the AO form, to the CPHF tolerance;
2. translational invariance: $\sum_B \partial^2 E / \partial A_k \partial B_l = 0$ for every A, k, l. Before symmetrizing (18), the sum over the first index x is exact, since $\sum_x B^x = 0$ and $\sum_x S^x = 0$ are integral identities; the sum over the second index y needs $\sum_y U^y = 0$ and so holds only to the CPHF tolerance (unless the last atom's U is taken as minus the sum of the others', section 3), and symmetrizing mixes the two;
3. rotational invariance at a converged stationary geometry: six zero eigenvalues (five for a linear molecule) after mass-weighting;

4. each term against PySCF separately (`partial_hess_elec` for the explicit part, `make_h1` for $F^{(x)}$, the `mo1` of `solve_mo1` for U), as the tree's gradient tests compare each term; for DF, against `pyscf.df.hessian.rhf` over the *same* aux set, not against the direct Hessian;

5. the whole against a central difference of `gradient::rhf` or `gradient::df_rhf`, which tests the tree against itself.

10 What to build, in order

1. **The semi-numerical Hessian.** A central difference of the existing analytic gradient: $6N$ SCF + gradient runs at 10^{-3} to 10^{-4} bohr steps, accurate to about 10^{-6} with tight SCF convergence. It needs no new kernel, gives frequencies now, and is the reference the analytic one is tested against. It belongs beside the geometry-optimization step as a pipeline step.

2. **Matrix-valued first derivatives.** S^x, h^x and the derivative J/K as $3N$ AO matrices (section 6), and the DF analogues. With these alone the CPHF right-hand sides exist.

3. **The batched CPHF solver** over a multi-density J/K build, testable on its own against `solve_mo1`.

4. **The response part of** E^{xy} (18): GEMMs and dot products once 2 and 3 exist. At this point the Hessian minus E^{xy}_{expl} can be checked against PySCF.

5. **Second-derivative integrals**, one-electron first (cheap, easy to verify), then the ERI or DF three- and two-centre families, generated by FelizJueves with translational invariance built in, and their compile times measured.

6. **Frequencies and intensities.** Mass-weighting, projection of translations and rotations, harmonic frequencies. IR intensities need the dipole derivatives, which reuse the U^x: $\partial \boldsymbol{\mu}_{\text{el}}/\partial x = -\operatorname{tr}(D\,\boldsymbol{\mu}^x) - 4\sum_{ai} U^x_{ai}\,\boldsymbol{\mu}_{ai} + 2\sum_{ij} S^x_{ij}\,\boldsymbol{\mu}_{ij}$ (with $\boldsymbol{\mu}$ the dipole integrals $\langle\mu|\mathbf{r}|\nu\rangle$, electrons carrying charge -1), plus the nuclear charge $Z_A\,\mathbf{e}_k$.

Steps 2–4 also serve any other second-order RHF property with a nuclear perturbation, and the batched CPHF serves electric ones (polarizabilities) with B replaced by the dipole integrals.

Since written. Step 1 is `src/pipeline/hessian`'s numerical method; step 6's dipole derivative and IR intensities are `hessian::rhf_dipole_derivative` and `hessian::infrared_intensities`, and its harmonic analysis is `geometry::harmonic_analysis`.

Part II
CASSCF

11 Scope, notation and the energy

Orbitals are real, the active space is complete, and the CI vector lives in a fixed determinant basis. MO indices: core i, j; active t, u, v, w; virtual a, b; general p, q, r, s. AO indices $\mu\nu\lambda\sigma$; aux indices P, Q. Nuclear coordinates are x and y, $3N$ of them.

With the 1- and 2-RDMs D_{pq} and Γ_{pqrs} over all MOs (core blocks closed-shell, active blocks from the CI vector), the energy is

$$E = \sum_{pq} h_{pq} D_{pq} + \tfrac{1}{2} \sum_{pqrs} (pq|rs)\, \Gamma_{pqrs} + V_{nn}, \tag{24}$$

and the generalized Fock matrix, `response::fock`'s, is

$$F_{pq} = \sum_{r} h_{pr} D_{rq} + \sum_{rst} (pr|st)\, \Gamma_{qrst}. \tag{25}$$

Only its first n_{occ} columns are nonzero. The parameters are $\lambda = (\kappa, c)$: the non-redundant rotations κ (core–active, core–virtual, active–virtual), $C(\kappa) = C\, e^{K}$ with K antisymmetric, and the CI vector. Their gradient is the orbital gradient $g_{pq} = 2(F_{pq} - F_{qp})$ and the CI gradient $2P(H-E)c$, $P = 1 - cc^{\mathsf{T}}$. Both vanish at convergence.

Two energies are in scope:

- **state-specific** (SS): one root, E stationary in all of λ (section 14);
- **state-averaged** (SA): root I's energy E_I at the orbitals that make $E_{\mathrm{SA}} = \sum_r w_r E_r$ stationary, with every root an eigenvector of the same active-space Hamiltonian H_0 (section 15).

Derivative integrals. As in Part I (section 1), h^x, $(\mu\nu|\lambda\sigma)^x$, S^x and their second derivatives are taken at fixed AO coefficients, and transformed to the MO basis with the unperturbed C: $S^x_{pq} = C_p^{\mathsf{T}} S^x C_q$. A quantity in brackets, $F^{[x]}$, $H^{[x]}$, $g^{[x]}$, means the same expression as F, H, g evaluated at the derivative MO integrals $(h^x_{pq}, (pq|rs)^x)$ in place of $(h_{pq}, (pq|rs))$, at the unperturbed RDMs.

12 The gradient, as the tree computes it

`mcscf::df_gradient` (`src/mcscf/gradient/README.md`) computes

$$E^x = \sum D^{\mathrm{AO}}_{\mu\nu} h^x_{\mu\nu} + \tfrac{1}{2} \sum \Gamma^{\mathrm{AO}}_{\mu\nu\lambda\sigma} (\mu\nu|\lambda\sigma)^x - \sum W_{\mu\nu} S^x_{\mu\nu} + V^x_{nn}, \qquad W = C\ \mathrm{sym}(F)\, C^{\mathsf{T}}, \tag{26}$$

with the connection $C(x) = C(1 - \frac{1}{2} S^x_{\mathrm{MO}}\, \delta x)$, so that $\partial E/\partial U_{pq} = 2F_{pq}$ turns the connection into $-\sum F_{pq} S^x_{pq}$. For a state average the same terms are taken at root I's Lagrangian densities: root I's RDMs, plus the transition RDMs of z_c, plus the orbital response of z_o (section 15).

The Hessian is d/dy of (26), or equivalently the second derivative of the energy along the parameter-fixed path plus the response. The second view is the one that organizes the work, and it starts with the path.

13 The connection, to second order

At fixed parameters the MOs must stay orthonormal as the basis moves. With $s(x) = C^{\mathsf{T}} S(x) C - 1$, the symmetric (Löwdin) connection is

$$C(x) = C\, T(x), \qquad T = (1+s)^{-1/2} = 1 - \tfrac{1}{2} s + \tfrac{3}{8} s^2 - \cdots \tag{27}$$

Expanding $s = S^x\, \delta x + \frac{1}{2} S^{xy}\, \delta x\, \delta y$ (MO basis throughout) gives the first and second derivatives of T,

$$\tau^x = -\tfrac{1}{2} S^x, \qquad \tau^{xy} = -\tfrac{1}{2} S^{xy} + \tfrac{3}{8} (S^x S^y + S^y S^x). \tag{28}$$

The tree's gradient uses τ^x only. **The linearized connection $1 - \frac{1}{2}S^x\delta x$ is not orthonormal at second order**, and a Hessian built on it is wrong by terms in S^xS^y. Any connection that is orthonormal to second order gives the same final Hessian at a stationary point; the pieces differ. The symmetric one is used here because it is the one the gradient and `response::fock` already assume.

The fixed-parameter second derivative. Write $E(T)$ for the energy (24) with every MO integral transformed by a general, not necessarily orthogonal, T: $h \to T^{\mathsf{T}}hT$, and T on each index of $(pq|rs)$. Then $\partial E/\partial T_{pq} = 2F_{pq}$ at $T = 1$, and its second derivative along two generators A and B is the energy at the *doubly one-index transformed* integrals,

$$E^{(2)}[A,B] = \sum D_{pq}(A^{\mathsf{T}}hB + B^{\mathsf{T}}hA)_{pq} + \tfrac{1}{2}\sum \Gamma_{pqrs}\,(pq|rs)^{[A,B]}, \tag{29}$$

$(pq|rs)^{[A,B]}$ being the sum over every way of applying A to one index and B to another. This is the machinery behind the orbital Hessian, `response::Hessian`, used here with symmetric generators instead of antisymmetric ones. Collecting terms, the second derivative of the energy at fixed λ is

$$\boxed{\begin{aligned} E_{xy} = &\underbrace{\sum D^{\mathrm{AO}}h^{xy} + \tfrac{1}{2}\sum \Gamma^{\mathrm{AO}}(\mu\nu|\lambda\sigma)^{xy} + V_{nn}^{xy}}_{\text{explicit}} + 2\sum_{pq} F_{pq}\,\tau_{pq}^{xy} \\ &+ 2\sum_{pq}\left(F_{pq}^{[x]}\,\tau_{pq}^{y} + F_{pq}^{[y]}\,\tau_{pq}^{x}\right) + E^{(2)}[\tau^x,\tau^y]. \end{aligned}} \tag{30}$$

The second term reduces to $-\operatorname{tr}(WS^{xy})$ plus the $\frac{3}{8}$ terms, the RHF $-WS^{xy}$ plus its second-order correction. The third is the derivative-integral Fock matrix against the first-order connection. The last is the connection acting twice. For RHF these collapse into the S_{ij}^x terms of the RHF Hessian's symmetric form.

14 State-specific CASSCF

14.1 The Hessian

$E(x,\lambda)$ is the energy at geometry x and parameters λ along the connection of section 13, with the orbitals

$$C(x,\kappa) = C\,T(x)\,e^{K(\kappa)}\,: \tag{31}$$

the connection first, the rotation second. The order changes the pieces, not the total. Every per-coordinate formula below, and the z-density shortcut of section 15, assumes this one. At the solution $E_\lambda = 0$ at every geometry, so

$$\frac{dE}{dx} = E_x, \qquad \frac{d^2E}{dx\,dy} = E_{xy} + E_{x\lambda}\,\lambda^y, \qquad E_{\lambda\lambda}\,\lambda^y = -E_{\lambda y}, \tag{32}$$

and therefore

$$\boxed{E^{xy} = E_{xy} - E_{x\lambda}\,E_{\lambda\lambda}^{-1}\,E_{\lambda y} = E_{xy} - \lambda^{x\mathsf{T}}E_{\lambda\lambda}\,\lambda^y.} \tag{33}$$

The response term is negative semi-definite when $E_{\lambda\lambda}$ is positive definite, a minimum in the parameters, as for RHF. A state-specific *excited* root is a saddle point in λ: the response term then has no sign, and a conjugate-gradient solve of $E_{\lambda\lambda}\lambda^y = -E_{\lambda y}$ is not guaranteed to converge. E_{xy} is (30). $E_{\lambda\lambda}$ is the CASSCF parameter Hessian (orbital–orbital, orbital–CI, CI–CI), the one the second-order optimizer and the Z-vector already apply. $E_{x\lambda}$ is the right-hand side below. The design checked (33) numerically against a central difference of PySCF's analytic CASSCF gradient.

14.2 The CP-CASSCF right-hand side

$E_{\lambda y}$ is the parameter gradient differentiated at fixed parameters along the connection:

$$\text{orbital:}\quad g^{[y]}_{pq} + (g \text{ at the integrals one-index transformed by } \tau^y)_{pq}, \qquad g^{[y]}_{pq} = 2(F^{[y]}_{pq} - F^{[y]}_{qp}), \tag{34}$$

$$\text{CI:}\quad 2P(\tilde{H}^y - \langle c|\tilde{H}^y|c\rangle)c, \qquad \tilde{H}^y = H^{[y]} + H[\tau^y], \tag{35}$$

where $H[\tau^y]$ is the active-space Hamiltonian at the τ^y one-index transformed integrals (core included, through h_{eff}). $F^{[y]}$ needs the derivative integrals with all four indices in the occupied and general spaces, contracted with D and Γ: a *derivative J/K build* over the core density and the active density, plus the active-pair derivative integrals $(pt|uv)^y$. These are the CASSCF version of the RHF skeleton Fock matrices (section 6), and like those they are matrices per coordinate, not the scalars the tree's derivative kernels return.

14.3 Solving it: the Z-vector's operator, $3N$ times

$E_{\lambda\lambda}\lambda^y = -E_{\lambda y}$ is the Z-vector equation with a different right-hand side. `response::zvector` (`src/response/zvector/README.md`) already solves it with a nonzero CI part, by the Schur complement on the CI-relaxed orbital Hessian $\tilde{H}$ of `response::RelaxedHessian`:

$$\tilde{H}\,\kappa^y = -g_y - H_{oc}w_y, \qquad w_y = -(H_0 - E)^{-1}P\,y_y, \qquad c^y = -(H_0 - E)^{-1}P\,H'(\kappa^y)\,c + w_y, \tag{36}$$

with (g_y, y_y) the orbital and CI parts of $E_{\lambda y}$. The scaling trap that README documents applies unchanged: its CI rows are *unweighted and unscaled*, so a right-hand side written with the $2P(H-E)c$ convention of (35) enters halved. Assemble the response term as $\Lambda^{\mathsf{T}}E_{\lambda\cdot}$, the solutions against the right-hand sides, not as $-\Lambda^{\mathsf{T}}E_{\lambda\lambda}\Lambda$. The operator's scaling then cancels between the solve and its right-hand side and never reaches the Hessian.

What changes is the count. One $\tilde{H}$ product is J/K builds plus an inner conjugate-gradient CI solve (up to 200 sigmas per root), so $3N$ right-hand sides through the Schur complement nest one iterative solve inside another, $3N$ times. Two remedies, to be measured rather than assumed:

1. **Batch the Schur complement.** All $3N$ share $\tilde{H}$, so block CG applies one product to $3N$ vectors: one batched J/K build and $3N$ CI solves per iteration.

2. **Solve the coupled system unreduced.** PCG on the full (κ, c) vector with the block-diagonal preconditioner ($\operatorname{diag} H_{oo}$, $(H_0 - E_0)_{\text{diag}}$), as PySCF solves its CP-MCSCF equations. No inner solve; each iteration is one orbital Hessian product, one $H'(\kappa)c$ and one sigma per right-hand side.

What the J/K builds see. Wang et al. count two J and two K builds per PCG step of TeraChem's CP-MCSCF, on two densities built from the trial vector p: the core–virtual density D_{CV} (only the $p_{\text{core,virt}}$ block) and the generalized active density

$$D_{\text{Gen}} = \begin{pmatrix} 0 & p_{\text{core,act}}(2-\gamma) & 2p_{\text{core,virt}} \\ 0 & 0 & \gamma\, p_{\text{act,virt}} \\ 0 & 0 & 0 \end{pmatrix}$$

(their Eqs. 14 and 15). Only the symmetric parts contribute, but the symmetrized densities are not positive semi-definite, so they cannot be written XX^{T} for DF-K-MO. Their fix is an SVD, $\tilde{D} = U\Sigma V^{\mathsf{T}}$, giving left and right pseudo-coefficients $C_L = U\Sigma^{1/2}$ and $C_R = V\Sigma^{1/2}$, and an exchange

build $K(C_L, C_R)$ over the fitted pairs. **The symmetrized $\tilde{D}$ has rank up to $2n_{\mathrm{occ}}$, not n_{occ}.** The unsymmetrized D_{Gen} and D_{CV} have nonzero rows only in the occupied block, so their rank is at most n_{occ} (or n_{core}). They also come factored, with no SVD: in the AO basis $D = C_o\,(D_o C^{\mathsf{T}})$, so $C_L = C_o$ and $C_R = C\,D_o^{\mathsf{T}}$. $K(C_L, C_R)$ followed by $K \to \frac{1}{2}(K + K^{\mathsf{T}})$ is then $K(\tilde{D})$ at half the rank. The SVD is a compression to try, not a requirement. Every density the Hessian adds is of this kind: the $3N$ response densities per CP-CASSCF iteration, the CI responses' transition RDMs, and in the state average the Z-weighted ones. A batched, left/right DF-K over SVD factors is the J/K the Hessian wants.

Since written. `response::zvectors` takes the $3N$ right-hand sides as a block, and solves them column by column, by the batched Schur complement, or by the coupled system unreduced (`src/response/zvector/README.md`, "Many right-hand sides").

15 State-averaged CASSCF

15.1 The Lagrangian, and why its second derivative needs only Z

Root I's energy is not stationary in the SA parameters. The gradient is that of

$$\mathcal{L}(x, \lambda, z) = E_I(x, \lambda) + z \cdot g_{\mathrm{SA}}(x, \lambda), \qquad g_{\mathrm{SA}} = \partial E_{\mathrm{SA}}/\partial\lambda, \tag{37}$$

with $Az = -\partial E_I/\partial\lambda$, $A = \partial g_{\mathrm{SA}}/\partial\lambda$ the SA Hessian: the tree's Z-vector. $\mathcal{L}$ is stationary in λ and z, and the same elimination as the NEVPT2 Hessian's $2n+1$ form (`docs/nevpt2/nevpt2.tex`, Eq. (hess)) gives

$$\boxed{E_I^{xy} = \mathcal{L}_{xy} + \mathcal{L}_{x\lambda}\lambda^y + \lambda^{x\mathsf{T}}\mathcal{L}_{\lambda y} + \lambda^{x\mathsf{T}}\mathcal{L}_{\lambda\lambda}\lambda^y, \qquad A\,\lambda^x = -\partial_x g_{\mathrm{SA}}.} \tag{38}$$

- λ^x is the response of the *SA* parameters: $3N$ solves with the SA Hessian A, the same operator as the Z-vector, against the SA right-hand sides (34)–(35) at the averaged densities.
- z enters at zeroth order, the gradient's own. No z^x, which would be $3N$ more solves.
- $\mathcal{L}_{xy} = E_{I,xy} + z \cdot \partial_{xy} g_{\mathrm{SA}}$: the fixed-parameter second derivative (30) at root I's densities plus the same at z's, the transition RDMs of z_c and the z_o-rotated densities. These are exactly the densities the gradient already forms.
- $\mathcal{L}_{x\lambda} = \partial_x\partial_\lambda E_I + z \cdot \partial_x\partial_\lambda g_{\mathrm{SA}}$: root I's CP right-hand side, plus the SA Hessian at derivative integrals (and connection) contracted with z.
- $\mathcal{L}_{\lambda\lambda} = \partial_{\lambda\lambda} E_I + z \cdot \partial_{\lambda\lambda} g_{\mathrm{SA}}$: root I's parameter Hessian, plus a **third derivative of E_{SA} contracted once with z.** Only its action $\lambda^{x\mathsf{T}}(\cdot)\lambda^y$ is needed.

The third-derivative term is the new object, and the warning the NEVPT2 Hessian carries applies to it verbatim: **differencing a Hessian-vector product does not give it** on any pair with an orbital part, because the operator is the Jacobian of g only at a stationary point. An analytic version differentiates `response::fock`'s gradient twice: J/K builds at doubly rotated densities, plus the CI legs as transition densities.

Since written. The third-derivative action is `mcscf::third_derivative` (`src/mcscf/third_derivative`), the covector $a \mapsto T(a, v, z)$ for each vector v of a block.

15.2 The rotations among the averaged roots

The SA energy is invariant to rotations among its roots when their weights are equal. Root I's is not. They are fixed by the condition that H_0 stays diagonal in the roots' span, $\langle c_r|H|c_s\rangle = 0$ for $r \neq s$. With $c_r \to c_r + \sum_s \theta_{sr} c_s$, θ antisymmetric, its first derivative gives

$$\theta^x_{sr} = \frac{\mathcal{H}^x_{rs}}{E_r - E_s}, \qquad \mathcal{H}^x_{rs} = \langle c_r|\tilde{H}^x + H'(\kappa^x)|c_s\rangle, \tag{39}$$

the orbital-relaxed coupling between the two roots, which is the derivative-coupling numerator. The CI response outside the span does not enter it, because $H_0 c_s = E_s c_s$ is orthogonal to that response.

In the gradient. At equal weights the rotations are redundant in E_{SA}, and E_I is stationary in them because the roots are eigenvectors. Their multipliers vanish, and the tree projects them away (`src/response/zvector/README.md`). **At unequal weights they are not redundant.** $\partial^2 E_{\mathrm{SA}}/\partial\theta^2_{rs} = 2(w_r - w_s)(E_s - E_r)$, and they couple to the orbitals through $(w_r - w_s)$ times the r–s transition orbital gradient, so they belong in A and can carry multipliers. A PySCF model of the state average finds:

- **two roots, any weights**: the multiplier of the one rotation is zero to 3×10^{-7} at weights $(0.7, 0.3)$, $(0.9, 0.1)$ and $(0.35, 0.65)$, against a coupling up to 0.16. The projected gradient is exact. This is observed, not proven;
- **three roots, weights** $(0.5, 0.3, 0.2)$: the rotations θ_{01} and θ_{02} again carry no multiplier, but θ_{12} carries 0.07–0.15. The projected gradient is **wrong by up to** 3.3×10^{-3}, 7% of root 2's gradient. With the rotations in A it is exact to 2×10^{-6}.

So the gradient as it then was is right at equal weights and for two roots, and wrong for three or more roots at unequal weights. That case is outside PySCF's analytic SA gradient too, which refuses unequal weights. Until the tree carries the rotations for pairs of unequal weight, it should refuse that case. Partly equal weights, where some rotations are redundant and some are not, were not checked.

In the Hessian. At equal weights the rotations are not in λ, and they add a term of their own to (38). E_I is stationary in θ_{sI}, with $\partial^2 E_I/\partial\theta^2_{sI} = 2(E_s - E_I)$ and total derivative $2\mathcal{H}^x_{Is}$ along the response. Eliminating θ then gives

$$\boxed{E^{xy}_I \mathrel{+}= \sum_{s\neq I} \frac{\mathcal{H}^x_{Is}\mathcal{H}^y_{sI} + \mathcal{H}^y_{Is}\mathcal{H}^x_{sI}}{E_I - E_s} = 2\sum_{s\neq I} \frac{\mathcal{H}^x_{Is}\mathcal{H}^y_{Is}}{E_I - E_s}.} \tag{40}$$

The factor 2 is not optional. With only one copy of the product, the Hessian is wrong by half this term, 0.02 to 0.24 in the design's checks. With three equal roots the term is as large as the Hessian itself. At unequal weights the rotations are part of λ, their response is inside λ^x, and (38) is complete without (40).

Either way the term **diverges as** $E_s \to E_I$. That is the physics: at a conical intersection the adiabatic energy is not differentiable and has no Hessian. `RelaxedHessian` already drops the other-root part of the CI response for a gap under 10^{-8} hartree. A Hessian must instead refuse, with the gap in the message, below a threshold on $|\mathcal{H}^x_{Is}|/|E_I - E_s|$ that is far larger. The tree's weights are recovered from a solution and need not be equal, so both paths are needed.

Since written. Every pair of unequal weight is a parameter of A with a multiplier of its own (`src/response/zvector/README.md`, "The rotations among the roots"), and the equal-weight term (40), with the rotations (39), is `mcscf::root_rotation` (`src/mcscf/root_rotation`).

16 The explicit part over density fitting

The explicit part of (30) is the second-derivative integrals contracted with D^{AO} and Γ^{AO}. One-electron: as RHF, at the CASSCF D. Two-electron, over a Coulomb-metric fit with $T_{a,P} = (a|P)$ for an AO pair $a = (\mu\nu)$, $N = M^{-1}$ and fitted pairs $c = TN$: for any symmetric pair-space density Γ_{ab} (the energy being $E_2 = \frac{1}{2}\sum_{ab}\Gamma_{ab}(TNT^{\mathsf{T}})_{ab}$),

$$\boxed{E_2^{xy} = \sum_{aP}\tilde{\Gamma}_{aP}\,T^{xy}_{aP} - \tfrac{1}{2}\sum_{PQ}\hat{W}_{PQ}\,M^{xy}_{PQ} + \sum_{ab}\Gamma_{ab}\big[R^x N R^{y\mathsf{T}}\big]_{ab}, \qquad R^x = T^x - c\,M^x,} \tag{41}$$

with $\tilde{\Gamma} = \Gamma c$ and $\hat{W} = c^{\mathsf{T}}\Gamma c$, the same intermediates as the gradient. For $\Gamma = D \otimes D$ this is Part I's DF Coulomb second derivative (21). The first two terms are the gradient's contractions with second-derivative integrals. The third is new, and it is affordable only because Γ factorizes. The tree's gradient already splits E_2 into the pieces that make it so (`src/mcscf/gradient/README.md`):

- **J**, core–core and core–active: $\Gamma = D \otimes D - D_a \otimes D_a$. The cross term is two Gram matrices of vectors $u^x - M^x d$, length n_{aux}, as for RHF.
- **K**, core with core and with active: $-\sum_i \sum_{pq} G_{pq}(ip|iq)$. The cross term needs R^x half-transformed to (i,p) pairs, $3N\,n_{\mathrm{core}}n_{\mathrm{occ}}n_{\mathrm{aux}}$ in all. Build it per atom.
- **active–active**: $\frac{1}{2}\sum\Gamma_{tuvw}(tu|vw)$. The cross term is $\sum\Gamma_{tuvw}(R^x_{tu})^{\mathsf{T}}NR^y_{vw}$ with R^x over active pairs, $3N\,n_{\mathrm{act}}^2 n_{\mathrm{aux}}$: small, and the cleanest of the three.

The left/right forms of Wang et al.'s SI (their Eqs. S9 and S15, the DF-J and DF-K gradients at independent D_L and D_R) are the gradients these cross terms differentiate. Their DF-K-MO variant (Eqs. S16–S19), which contracts over MO pairs and never forms an $n_{\mathrm{aux}}n_{\mathrm{AO}}^2$ transition tensor, is the right shape for the R^x half-transforms too.

The connection terms of (30) need $F^{[x]}$ (the derivative J/K builds of section 14) and $E^{(2)}[\tau^x, \tau^y]$, which is the orbital-Hessian contraction with symmetric generators. Where τ^x and τ^y fall on different pairs of a core density, it is J/K builds on $C\tau^x$-rotated densities, $3N$ of them, contracted pairwise. Where both fall on the same pair, the core part is $\tau^{x\mathsf{T}}J(D)\tau^y$-type contractions of the unperturbed J/K. Where they meet the active 2-RDM, it needs the $(pq|tu)$- and $(pt|qu)$-type integrals with two general indices: the `ppaa` and `papa` blocks that `response::Hessian` already contracts for antisymmetric generators.

Since written. (41), split as above, is `hessian::df_casscf_two_electron`, and (30) built on it is `hessian::df_casscf_exy`; $F^{[x]}$ and the right-hand sides (34)–(35) are `mcscf::df_derivatives` (`src/mcscf/derivative`).

17 The algorithm

The state average adds, in order of difficulty: the z-weighted copies of lines 1–4 (the gradient's densities, one more pass); the root-rotation responses (39), closed form once λ^x is known; the z-contracted SA Hessian at derivative integrals in $\mathcal{L}_{x\lambda}$; and the third-derivative action $\lambda^{x\mathsf{T}}(z\cdot\partial_{\lambda\lambda}g_{\mathrm{SA}})\lambda^y$,

Algorithm 2 State-specific CASSCF analytic Hessian, Eq. (33)

Input: converged C, c; D, Γ, F; basis and aux basis
Output: E^{xy}, $3N \times 3N$

$H \leftarrow$ explicit part: second-derivative integrals against D^{AO}, Γ split as in section 16; $+V_{nn}^{xy}$
for each coordinate x (batched) **do**
    S^x, h^x, derivative J/K at the core and active densities, $(pt|uv)^x$ ▷ matrix-valued derivatives
    $\tau^x \leftarrow -\frac{1}{2}S^x$; $F^{[x]}$; $E_{\lambda x}$ by (34)–(35)
$H \mathrel{+}= 2F \cdot \tau^{xy} + 2(F^{[x]} \cdot \tau^y + F^{[y]} \cdot \tau^x) + E^{(2)}[\tau^x, \tau^y]$ ▷ connection, Eq. (30)
solve $E_{\lambda\lambda}\lambda^x = -E_{\lambda x}$ for all x: block PCG, batched products ▷ the Z-vector operator
$H \mathrel{+}= -\Lambda^{\mathsf{T}} E_{\lambda\lambda} \Lambda \;=\; \Lambda^{\mathsf{T}} E_{\lambda\cdot}$, $\Lambda = (\lambda^1 \cdots \lambda^{3N})$
check symmetry and the translational sum rule; symmetrize

$3N(3N+1)/2$ pairs, each a doubly-rotated J/K build unless it is organized as $3N$ actions on λ^y followed by dot products.

Cost. As for RHF, the response dominates: $3N$ CP-CASSCF solves, each costing about one Z-vector, against a gradient that costs one. A CASSCF Hessian is therefore about $3N$ CASSCF gradients plus the explicit part, and the semi-numerical route costs $6N$ CASSCF runs *plus* $6N$ gradients. The analytic route wins by more than the factor of two that counting suggests, because each displaced CASSCF re-converges the orbitals from scratch. That is an expectation from the operation counts, not a measurement.

Since written. `hessian::df_casscf` is Algorithm 2, with its CP-CASSCF step on its own as `hessian::df_casscf_cp`; the state average is `hessian::df_sacasscf`, its z terms `hessian::df_sacasscf_z` (`src/hessian/README.md`).

18 Differentiability

Every condition a smooth CASSCF gradient needs, the Hessian needs too, and more sharply, because the response now enters quadratically:

- a non-singular $E_{\lambda\lambda}$ (or A), a true minimum in the orbital–CI space. A near-singular one, such as an active orbital with occupation near 0 or 2 that is nearly redundant with the core or virtual space, makes λ^x large and the Hessian ill-conditioned before it makes the gradient wrong;
- no root flipping in the state average and no active-space swap along the displacements. The semi-numerical Hessian is exposed to this, the analytic one is not, and the disagreement between them is a diagnostic;
- for the state average, a gap to every other averaged root (section 15.2).

19 Checks

The state-specific formula. The design's check evaluates the CASSCF energy $E(x, \lambda)$ directly: Löwdin connection to all orders, orbitals $C(x)e^{K(\kappa)}$, and the CI vector $c_0 + Q\theta$ normalized, with Q an orthonormal basis of c_0's complement. It takes E_{xy}, $E_{x\lambda}$ and $E_{\lambda\lambda}$ by central finite differences

at fixed parameters and assembles (33). The reference is a central difference of PySCF's analytic CASSCF gradient, re-converged at each displaced geometry. For water, CAS(4,4)/6-31G, at a C_1 geometry with both O–H bonds stretched (1.40 and 1.51 Å), so that the active space is the two σ/σ^* pairs with natural occupations near 1.88, 1.84, 0.16 and 0.12:

comparison	max. abs. difference (E_h/a_0^2)
formula (33) vs. FD of PySCF's analytic gradient	2.8×10^{-4}
asymmetry of that FD reference (its own noise)	2.3×10^{-4}
diagonal of (33) vs. FD of self-relaxed energies	2.5×10^{-6}
E_{xy} alone (no response) vs. FD of PySCF's gradient	4.8×10^{-1}
asymmetry of (33)	2×10^{-16}

The largest Hessian element is 0.06, and the response term is not small: dropping it misses by 0.48. The first row sits at the reference's own noise, set by PySCF's reachable gradient convergence (10^{-6}). The third row replaces PySCF with a reference that has no solver noise: $E(x, \lambda)$ relaxed to 10^{-9} in its gradient at each displaced geometry by Newton steps, the relaxed energies then differenced. This checks the structure (33) and the connection, but not the analytic expansions (30)–(35) term by term.

The analytic terms. A second check takes the derivative AO integrals (first and second, at fixed AO coefficients) by finite difference. It assembles E_{xy} by (30) and $E_{\lambda x}$ by (34)–(35) exactly as written, with $E^{(2)}$ over every pair of index positions. It compares them with the fixed-parameter finite differences above, at the same molecule:

term	max. size	max. abs. difference
E_{xy}, (30)	0.44	2.5×10^{-7}
the same without the $\frac{3}{8}(S^xS^y + S^yS^x)$ of τ^{xy}		9.3×10^{-1}
orbital right-hand side, (34)	2.0	1.5×10^{-6}
CI right-hand side, (35)	0.21	2.5×10^{-7}

The second-order connection is not a small correction: without its S^xS^y terms E_{xy} is wrong by twice its own size.

The state average. A third check does the same for SA-CASSCF(4,4) over two or three singlet roots, at the same geometry. The roots are $(Ve^A)_{\cdot r}$ over an orthonormal basis V of the singlet CI space, A holding the root–complement rotations and, as separate parameters, the root–root rotations θ. The reference solution is polished by Newton steps to $|g_{\mathrm{SA}}| < 10^{-10}$. The reference energies are re-relaxed at each displaced geometry and the roots re-diagonalized in their span. Every piece of (38) is a finite difference of the energy matrix $\langle c_r|H|c_s\rangle$, with the z terms as a directional difference of E_{SA} along z. The Hessian is compared on its diagonal and along three random directions. The gradient compares the Lagrangian with the rotations projected (the tree, PySCF) and with them in A against the differenced relaxed energies. Each row is the worst over the roots:

weights	gradient, projected	gradient, rotations in A	Hessian
$(0.5, 0.5)$	6×10^{-7}	—	3×10^{-5}
$(0.7, 0.3)$	6×10^{-7}	6×10^{-7}	5×10^{-5}
$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	2×10^{-6}	—	2×10^{-5}
$(0.5, 0.3, 0.2)$	$\mathbf{3.3 \times 10^{-3}}$	2×10^{-6}	3×10^{-5}

The Hessian column is (38), plus (40) at equal weights. The largest Hessian elements are 0.05 to 0.48 and the largest gradients 0.04 to 0.16. Without (40) the equal-weight Hessians miss by 0.04 (two roots) and 0.48 (three). With one copy of the product instead of two they miss by half that. The residuals are round-off of the nested differences, not a missing term: they grow about tenfold for each halving of the step (2×10^{-3} here). Section 15.2 has what the gradient rows mean.

$\mathcal{L}_{x\lambda}$ is also checked analytically in x, both parts against the mixed (x, λ) difference, at every weight set above: root I's right-hand side (34)–(35) at its own densities (the root–root rotations included at unequal weights) to 6×10^{-6} of a maximum of 1.9, and z times the SA Hessian at the derivative integrals plus the τ^x one-index transform to 5×10^{-6} of a maximum of 0.14. The SA Hessian product in the second is still a difference in λ; it is the operator the tree already has. Not checked analytically: the z part of $\mathcal{L}_{xy}$ ((30) at z's densities), the third derivative, and partly equal weights.

Density fitting. A fourth check repeats the state-specific checks over a Coulomb-metric fit (cc-pVDZ-JKFIT), at the same molecule. The DF first-derivative four-index integrals are formed analytically, $T^x c^\mathsf{T} + cT^{x\mathsf{T}} - cM^x c^\mathsf{T}$, from finite-difference T^x and M^x. PySCF's DF-CASSCF is only the starting point. The solution is polished on the check's own DF energy, and the reference is that energy relaxed by Newton steps on its analytic gradient to 10^{-10}:

comparison	max. abs. difference
(41) vs. FD of $\frac{1}{2}\Gamma \cdot (TNT^\mathsf{T})$, fixed Γ^{AO}	3.4×10^{-8}
the same without the $R^x N R^{y\mathsf{T}}$ term	3.2×10^{-2}
J + K + active cross terms (section 16) vs. generic Γ_{ab}	5×10^{-14}
E_{xy}, (30), over DF integrals	2.1×10^{-7}
orbital and CI right-hand sides	1.5×10^{-6}, 2.4×10^{-7}
(33) vs. relaxed DF energies (diagonal, 3 directions)	1.0×10^{-6}

The metric is ill-conditioned, $\kappa(M) \approx 2 \times 10^6$ here. An explicit M^{-1} puts 10^{-10} of round-off on every energy, which a second difference at step 10^{-3} turns into 10^{-4}: the first run of the check missed by that much. Contract through the Cholesky factor of M in anything that is differenced, the semi-numerical Hessian of section 20 included. The analytic Hessian never differences an energy and is not affected.

For the GPU implementation.

1. symmetry, and the translational sum rule;
2. E_{xy} alone against a finite difference of the energy at fixed parameters, exactly as the check does: it isolates the connection, the bug most likely to survive everything else;
3. each λ^x against the difference of converged parameters at displaced geometries (after aligning phases and the connection);
4. the whole against a central difference of `mcscf::df_gradient`, SS and each SA root. Include an equal-weight case with three roots, where (40) is as large as the Hessian, and an unequal-weight case;
5. against OpenMolcas, which computes analytic CASSCF Hessians, over the same auxiliary basis for the DF form.

Since written. `tools/emit/casscf_hessian_reference.py` ports these checks onto density fitting throughout, and its data under `test/gtest/data/casscf_hessian/` reproduce the tables above.

20 What to build, in order

1. **The semi-numerical Hessian** over `mcscf::df_gradient`, SS and SA, with the previous geometry's orbitals and CI as guess and a check that the active orbitals and the root order did not change. It is the reference everything below is tested against, and it is only as good as its displaced solves. Building this document's check hit two failures that raised no error and gave differenced energies and gradients wrong by an order of magnitude. A weakly correlated CAS(4,4) near equilibrium never converged. Then a displaced CASSCF started from the reference orbitals, which are not orthonormal in the displaced overlap metric, converged to an energy $O(\delta x)$ off: PySCF does not re-orthonormalize a guess. Orthonormalize the guess with the connection, $C(S_{\mathrm{MO}})^{-1/2}$. Refuse an unconverged displacement.

2. **The RHF Hessian's kernels** (Part I, section 10, items 2, 3 and 5): matrix-valued first derivatives, the multi-density J/K, second-derivative integrals. The CASSCF Hessian needs all of them, at more densities.

3. **The left/right SVD-factored DF-K**, after Wang et al., for the non-PSD response and transition densities.

4. **The second-order connection** and E_{xy} (30), checked against fixed-parameter finite differences.

5. **The batched CP-CASSCF solve**, the Z-vector's operator on $3N$ right-hand sides. Measure the batched Schur complement against the unreduced coupled PCG before choosing.

6. **The SS Hessian** (33). Frequencies of a single-state CASSCF.

7. **The SA Hessian** (38): the z terms, the root rotations ((40) at equal weights, in λ otherwise) with their refusal, and last the third derivative. The NEVPT2 Hessian needs the same third derivative, so it is built once for both. First, and independently of the Hessian, make `mcscf::df_gradient` refuse three or more roots at unequal weights, or carry their rotations in the Z-vector (section 15.2).

Since written. Item 1 is `src/pipeline/hessian`'s numerical method over the CASSCF chain, which orthonormalizes its guess in the displaced metric and refuses a displaced solve that left the reference's solution (`src/pipeline/hessian/README.md`); items 6 and 7 are the modules of section 17's "Since written"; and `mcscf::df_gradient` carries the unequal-weight rotations.

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